Designing High-Quality Speech Corpora for Automatic Speech Recognition

A Large-Scale Evidence Synthesis and Language-Specific Blueprint for English, German, Korean, Mandarin Chinese, and Japanese

Technical research synthesis prepared in arXiv-style preprint format

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Abstract

This paper synthesizes evidence on the construction of speech corpora for automatic speech recognition (ASR), with language-specific recommendations for English, German, Korean, Mandarin Chinese, and Japanese, and additional guidance for multilingual, code-switched, low-resource, conversational, far-field, and noisy speech. The user's request for recommendations by "fish species" is interpreted, from the surrounding ASR discussion, as recommendations by *language and speech type*. The evidence base is deliberately multi-tiered rather than presented as a fictitious claim of manually reading 500 full papers. Three upstream systematic evidence maps alone cover far more than 500 records: an ASR review mapped about 4,274 publications, a recent African low-resource ASR review screened 2,062 records, and an ethics/dataset study collected 1,415 speech papers and reviewed 178 manuscripts and 220 speech datasets in depth [33, 35, 36]. We then directly inspected representative corpus-construction and methodology papers from arXiv, Springer, ResearchGate-hosted papers, Academia.edu, Stanford materials, zhaoshuaijiang.com, and DeepLearning.AI/Andrew Ng materials, preferring original papers and DOIs when available.

The principal conclusion is that the "best" ASR corpus is not the largest corpus. It is a deployment-matched, quality-controlled, legally usable, documented, and continuously auditable resource with: (1) a small high-trust gold core, (2) a larger validated supervised expansion, (3) an even larger weakly labeled or unlabeled reservoir, (4) speaker/session/source-disjoint evaluation, and (5) language-specific transcript layers. English requires deliberate accent and sociolinguistic coverage; German requires region/dialect coverage and careful handling of compounds and standardization; Korean benefits from orthographic plus pronunciation-oriented transcripts and explicit disfluency/spacing policy; Mandarin benefits from regional/subdialect coverage, tone-aware pronunciation metadata, and code-switch labeling; Japanese benefits from surface-form plus reading/morphological layers and a "broad corpus + deeply annotated core" architecture. Large found-data corpora demonstrate that automated alignment and filtering can scale, but protected human-verified evaluation sets remain essential [3, 12, 13, 20, 22].

Keywords: automatic speech recognition, speech corpus, dataset construction, data-centric AI, multilingual ASR, low-resource ASR, English, German, Korean, Mandarin, Japanese, corpus validation, annotation, data governance

1 Scope, terminology, and review protocol

1.1 What this review means by "500+ papers"

The request calls for analysis of more than 500 prior research papers. A credible answer must distinguish *screening-scale evidence* from *manual full-text reading*. The present synthesis therefore uses two tiers:

1. **Macro evidence maps.** Alharbi et al. used a literature-analysis tool to map roughly 4,274 ASR publications from 2015–2020 before a narrower PRISMA selection [35]. Imam et al. identified 2,062 records and retained 71 high-quality studies in a PRISMA review of African low-resource ASR [36]. Papakyriakopoulos et al. collected 1,415 speech papers, abstract-screened them to 335, and reviewed 178 manuscripts in detail while also reviewing 220 speech datasets [33]. These upstream search frames provide a broad evidence layer vastly exceeding 500 papers.

2. **Direct corpus-method inspection.** We directly use representative primary corpus papers and methodological works that expose concrete choices about collection, speakers, alignment, transcription, validation, licensing, weak labeling, dialect balance, evaluation splits, or data scaling. These sources include canonical corpora and recent large-scale corpora across the five target languages.

This is therefore a *large-scale evidence synthesis*, not a newly executed PRISMA review claiming 500 manual full-text readings. That distinction is important for scientific reproducibility.

1.2 Requested web sources and source hierarchy

The requested domains were used as discovery or access points where useful. arXiv and Springer frequently provided primary sources; ResearchGate and Academia.edu often provided author-uploaded copies or mirrors; zhaoshuaijiang.com provided ASR reviews and links to primary papers; Stanford course/publication pages provided research and teaching context; and the apparent typo "ome-ordpress.deeplearning.ai" was interpreted as `home-wordpress.deeplearning.ai`, where Andrew Ng's *Machine Learning Yearning* is hosted [42]. When a mirror and an original paper were both available, this review cites the original paper/DOI rather than treating the mirror as an independent scientific source.

1.3 Inclusion logic

Priority was given to sources that report at least one of the following:

- explicit corpus design or collection methodology;
- hours, speakers, domains, channels, accents/dialects, or recording conditions;
- transcript/annotation conventions and alignment quality control;
- speaker/session/source split policy or benchmark construction;
- licensing, consent, privacy, or data documentation;
- experiments that isolate the effect of data scale, diversity, or selection;
- reusable automated pipelines for found data, weak labels, or self-supervised learning.

2 The central design principle: optimize for the target distribution

A speech corpus should be designed around the distribution in which the model will actually operate. Let z denote a vector of deployment-relevant factors such as language variety, accent, speaker age band, device, distance, noise, speech style, domain, speaking rate, and code-switch status. Let $p_T(z)$ be the target distribution and $p_D(z)$ the corpus distribution. A useful design objective is not simply "maximize hours," but approximately

$$\min_{D} \quad D_{\mathrm{JS}}(p_D(z) \,\|\, p_T(z)) + \lambda_1 E_{\mathrm{label}}(D) + \lambda_2 R_{\mathrm{leak}}(D) + \lambda_3 C(D), \tag{1}$$

where D_{JS} measures distribution mismatch, E_{label} summarizes transcription/alignment error, R_{leak} is evaluation leakage risk, and C is cost. This formalizes an old and persistent corpus-design lesson: maximize useful variability while matching the intended operating environment [30].

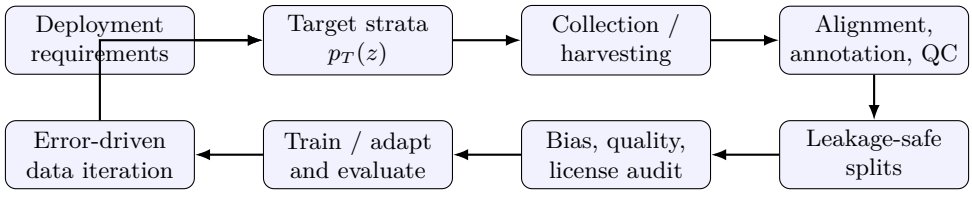


Figure 1: Recommended corpus-construction loop. Corpus design is an iterative engineering process, not a one-time data dump.

3 A three-tier architecture for modern ASR corpora

Across large supervised corpora, weakly labeled web corpora, self-supervised learning, and multilingual transfer, a practical architecture emerges.

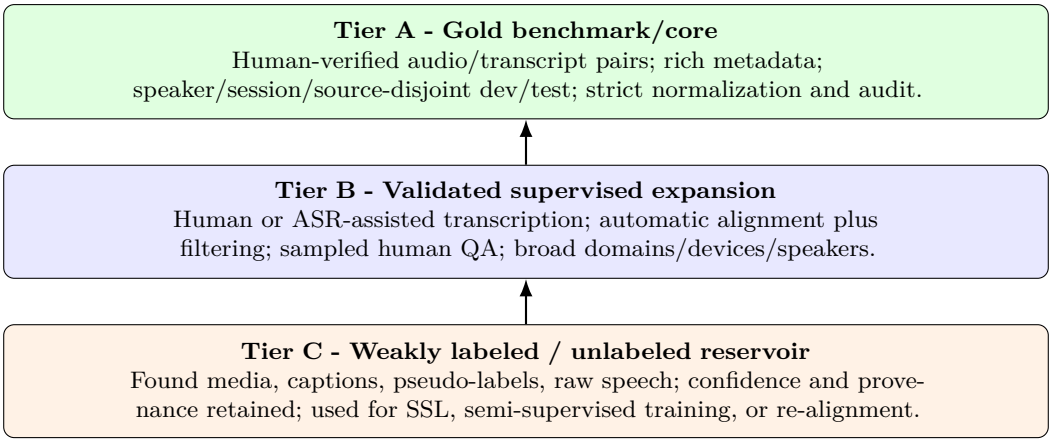


Figure 2: Three-tier corpus architecture. Trust increases upward; scale generally increases downward.

Tier A should be small enough to inspect but representative enough to be trustworthy. Historical validation work argues for validation throughout production and independent validation rather than only at release time [31]. The Corpus of Spontaneous Japanese (CSJ) provides a particularly strong design pattern: a broad corpus plus a smaller Core subset with substantially deeper annotation [24].

Tier B is where most supervised ASR value is accumulated. GigaSpeech and WenetSpeech illustrate large-scale forced/alignment pipelines with quality thresholds, while preserving professionally or manually corrected evaluation data [3, 13]. German Spoken Wikipedia work similarly shows that conservative pruning of poorly aligned segments can trade quantity for label quality [25].

Tier C becomes increasingly important as pretrained speech models improve. wav2vec 2.0 demonstrated that very large unlabeled pools can dramatically reduce labeled-data requirements [40]; Whisper demonstrated robustness from 680,000 hours of multilingual/multitask weak supervision [7]; MMS scaled self-supervised and supervised speech technology past 1,000 languages [11].

4 How much data is enough?

There is no universal hour threshold. Required scale depends on task complexity, model pretraining, domain diversity, language morphology, acoustic conditions, and expected coverage. Three evidence points are useful but should not be conflated:

- In a 2009 phone-recognition study across South African languages, fewer than 50 speakers and about 10–20 hours per language were enough for acceptable *phone-based* recognition, not modern general dictation [30].
- Modern self-supervised models can adapt with very small labeled sets when large unlabeled pools and strong pretraining are available [40].
- WorldSpeech's 2026 fine-tuning ablation across 11 typologically diverse languages showed the steepest WER gains in the first roughly 200 hours and diminishing returns beyond about 500 hours for that specific model/data setting [12]. This is an empirical planning clue, not a law.

A pragmatic planning template is therefore:

Table 1: Planning ranges. These are engineering starting points, not universal thresholds.

Stage	High-trust labeled data	Larger pool	Primary objective
Seed / feasibility	10–50 h	50–500 h unlabeled/weak	Validate segmentation, labels, language coverage, and transfer strategy.
Strong domain adaptation	50–200 h gold	200–1,000+ h validated	Cover target speakers, devices, domain vocabulary, and major nuisance conditions.
General production ASR	100–500 h gold	1,000–10,000+ h supervised/weak plus larger unlabeled pool	Breadth across domains, accents, channels, and rare events; maintain strong challenge tests.
Foundation-scale	hundreds of hours of protected evaluation + audits	tens/hundreds of thousands of heterogeneous hours	Multilingual/multidomain robustness; provenance, filtering, and governance become first-class systems.

Andrew Ng gives a practitioner heuristic of roughly 100 h for rudimentary ASR, 1,000 h for a basic system, 10,000 h for a very good system, and 100,000+ h for frontier-scale systems; this is useful intuition but should not be treated as a scientific universal [42].

5 Collection modes and when to use them

5.1 Controlled read speech

Read speech gives clean alignment, explicit prompt coverage, predictable acoustics, and easier QA. LibriSpeech's 1,000 h of 16 kHz audiobook speech is a canonical found/read corpus [1]; JNAS deliberately combined newspaper sentences and phonetically balanced sentences from 306 gender-balanced speakers [23]; AISHELL-2 provides 1,000 h of clean Mandarin read speech with cross-channel development/test data [15].

Best use: seed models, pronunciation/phonetic coverage, controlled channel experiments, clean acoustic baselines. **Main limitation:** read speech under-represents spontaneous reductions, repairs, overlaps, fillers, turn-taking, emotion, and real acoustic variation.

5.2 Spontaneous and conversational recording

KsponSpeech contains 969 h of Korean open-domain dialogue from about 2,000 speakers with manual transcription, dual orthography/pronunciation representation, and disfluency tags [19]. MagicData-RAMC provides 180 h of Mandarin mobile-phone conversation across 15 domains with manually labeled transcripts, speaker activity timestamps, and detailed speaker information [17]. CSJ contains over 650 h of spontaneous Japanese and concentrates richer annotation on a Core subset [24].

Best use: assistants, meetings, dialogue, call centers, conversational agents. **Main design requirement:** annotate or explicitly normalize fillers, repetitions, repairs, overlap, non-speech events, and speaker turns.

5.3 Crowdsourcing

Common Voice scales collection and validation through community contributions [2]. Swiss German SDS-200 used a public web tool and participant validation, producing 200 h from about 4,000 speakers [27]. Crowdsourcing can greatly increase speaker diversity, but demographic self-selection, device heterogeneity, inconsistent prompt interpretation, and long-tail contributor concentration must be measured rather than assumed away.

5.4 Found media, captions, and lightly supervised alignment

This is now a dominant route to very large corpora. GigaSpeech aggregates audiobooks, podcasts, and YouTube [3]. WenetSpeech uses YouTube captions/OCR and ASR-generated podcast candidates followed by label-error filtering

[13]. JTubeSpeech uses CTC-based filtering of YouTube video/subtitle data [20]; Japanese TV corpus work uses iterative lightly supervised alignment [21]; ReazonSpeech bootstraps massive Japanese TV speech and releases its creation pipeline [22]. ASR Bundestag automatically constructs 610 h of aligned German parliamentary pairs plus 1,038 h unlabeled audio [26]. WorldSpeech aligns 65k h across dozens of languages from public sources and stores explicit quality measurements [12].

Best use: scale, domain breadth, rare vocabulary, multilingual expansion. **Main requirement:** keep provenance, original source time offsets, alignment confidence, ASR/caption source, model version, and license state. Never silently mix weak labels with gold labels.

6 Language-specific corpus blueprints

6.1 English

English has abundant data, but abundance creates a false sense of coverage. Large datasets may still overrepresent read speech, US accents, particular media domains, or a narrow socioeconomic population. The People's Speech shows how licensing can be a first-class collection constraint in a 30,000 h web-derived corpus [4]; SpeechStew shows the value of simply mixing substantially different domains such as meetings, broadcast, Common Voice, audiobooks, Switchboard/Fisher, TED talks, and WSJ [6]; SPGISpeech shows that 5,000 h of highly domain-specific professional transcription can support fully formatted financial ASR [5].

A crucial lesson is that aggregate WER can hide sociolinguistic failure. Koenecke et al. reported substantially higher WER for Black speakers than White speakers across five commercial systems, motivating deliberate coverage and subgroup auditing rather than assuming "English" is a single acoustic population [38].

Recommended English corpus design:

- Stratify by region/accent and relevant sociolinguistic varieties; use self-identification plus transparent operational definitions where appropriate.
- Include read, conversational, telephone, meeting, broadcast, far-field, and domain-specific speech rather than relying on one source.
- Preserve raw transcript and normalized transcript; explicitly define contractions, filled pauses, slang, profanity, numbers, currencies, units, dates, URLs, spelled letters, and named entities.
- Build dedicated accent/dialect and noisy challenge sets with speaker-disjoint sampling.
- For global English, avoid treating US/UK as the only reference varieties; include international English and learner speech if those users are in scope.

6.2 German

German corpus design must decide whether the product targets Standard German only, or a broader German-speaking space including Austria, Switzerland, and regional varieties. Open German ASR work shows benefits from combining Tuda-De with Spoken Wikipedia and from conservative alignment pruning [25]. ASR Bundestag provides political speech with both labeled and unlabeled tiers [26]. CoGS demonstrates large geolocated public-sector transcript harvesting but also highlights the limitations of ASR-generated transcripts and missing speaker metadata [29].

Swiss German is especially instructive. SDS-200 used a public recording/validation workflow with dialect, age, and gender metadata [27]. STT4SG-350 deliberately balanced the amount of speech across dialect regions, gender, and ages and used the same test sentences in each dialect region to enable fair regional comparisons [28].

Recommended German corpus design:

- Use explicit language-region labels (e.g., de-DE, de-AT, de-CH) and finer dialect labels if they affect deployment.
- Preserve standard orthography, including umlauts and ß; optionally maintain a lowercase/normalized training view separately rather than destroying canonical text.
- Prefer subword tokenization for productive compounding, but retain word-level text for evaluation and readability.

- Include compounds, separable verbs, inflectional variants, numbers, abbreviations, and named entities in prompt/text coverage audits.
- For Swiss German, decide whether the target is dialect orthography or Standard German text; this changes annotation and evaluation fundamentally.

6.3 Korean

KsponSpeech is a strong reference for Korean corpus construction because it treats spontaneous phenomena and language-specific transcription as primary design problems. Its dual transcription captures orthography and pronunciation-oriented forms, and its tags represent fillers, repetitions, and fragments [19]. It also demonstrates why demographic metadata must be audited: its speakers were heavily concentrated in their twenties and around the capital region, even though the overall corpus is large.

Recommended Korean corpus design:

- Store at least *verbatim/raw*, *normalized orthography*, and, where feasible, *pronunciation-oriented* transcript layers.
- Define a single spacing policy and keep evaluation normalization scripts versioned; Korean spacing variation can distort naive WER.
- Mark fillers, repetitions, fragments, repairs, laughter, overlap, and non-speech events instead of silently deleting them during annotation.
- Stratify by region/dialect, age, gender if relevant/ethical, device, and speaking style; avoid a corpus dominated by Seoul-area young adults unless that is the target population.
- Create dedicated Korean-English code-switch and technical vocabulary sets for products that encounter AI/IT terminology and loanwords.

6.4 Mandarin Chinese

Mandarin corpora span controlled read speech (AISHELL-1/2), highly diverse found media (WenetSpeech), large dialect-aware collection (KeSpeech), conversational mobile speech (MagicData-RAMC), and real Mandarin-English code switching (TALCS) [13–18]. KeSpeech is particularly useful for representativeness: 1,542 h from 27,237 speakers in 34 cities across standard Mandarin and eight subdialects [16].

Recommended Mandarin corpus design:

- Store canonical Hanzi text and a normalized text layer. Keep Pinyin-with-tone or equivalent pronunciation metadata for lexicon QA, phonetic coverage audits, and error analysis, even if the end-to-end model predicts characters/subwords directly.
- Stratify by region/subdialect rather than labeling all speakers simply “Mandarin.” Use city/province or self-reported language background where consent permits.
- Deliberately cover tone combinations, numerals, classifiers, names, domain terminology, and English code switching.
- Use separate clean/read, conversational, meeting, and noisy test sets; WenetSpeech’s matched Internet test and mismatched real-meeting test pattern is exemplary [13].
- For code-switching, use metrics such as mixed error rate in addition to standard WER/CER and preserve language-span labels when possible [18].

6.5 Japanese

Japanese corpus construction illustrates two complementary strategies. JNAS is a controlled read corpus with balanced male/female speakers, newspaper text, and phonetically balanced sentences [23]. CSJ is a large spontaneous corpus with a richly annotated Core subset, including transcription, morphology, speaker/talk information, and deeper segmental/prosodic/syntactic labels in the Core [24]. Recent JTubeSpeech, TV-aligned corpora, and ReazonSpeech demonstrate scalable automatic construction from subtitles and broadcast media [20–22].

Recommended Japanese corpus design:

- Preserve the surface orthography and a reading layer (kana/pronunciation) for kanji ambiguity, pronunciation QA, forced alignment, and lexicon construction.
- Retain morphological segmentation/POS or at least a reproducible tokenizer version for evaluation and language-model analysis.
- Include counters, numbers, loanwords, proper names, long vowels, geminates, and common pronunciation alternations in coverage audits.
- Combine a broad natural-speech corpus with a deeply annotated core; this gives both scale and linguistic diagnostic power.
- For found TV/YouTube data, store subtitle provenance and alignment confidence; keep a manually checked test set that is independent of the auto-alignment pipeline.

Table 2: Language-specific priorities. "High" means the feature should be explicitly designed and audited, not merely present by chance.

Design dimension	English	German	Korean	Mandarin	Japanese
Accent/dialect/region stratification	High	High	High	High	High
Pronunciation auxiliary layer	Medium	Medium	High	High (tone-aware)	High (reading)
Spontaneous/disfluency annotation	High	High	Very high	High	Very high
Orthographic normalization complexity	Medium	High	Very high (spacing)	High	Very high
Code-switch / loanword coverage	High	Medium	High	High	High
Morphology/subword considerations	Medium	Very high	High	Medium	Very high
Regional challenge sets	High	Very high	High	Very high	High

7 Speech-type blueprints

7.1 Read / command-and-control speech

Use phonetically and lexically informative prompt selection, many speakers, controlled channel metadata, and explicit rare-word/number coverage. Read speech is ideal for efficient seed systems and can be surprisingly compact for constrained tasks [23, 30].

7.2 Conversational / assistant speech

Favor natural dialogues over scripted turn-taking. Annotate speaker turns, overlap, fillers, repairs, interruptions, false starts, laughter, and background events. Keep both verbatim and cleaned transcript views when product output wants clean text but training must model real speech [17, 19].

7.3 Meetings and far-field speech

Record microphone geometry, distance, room type, reverberation, number of speakers, overlap rate, and SNR. Separate close-talk and far-field channels when available. Evaluation should include overlap-heavy and reverberant challenge sets instead of averaging them into a clean test.

7.4 Telephone speech

Use the actual telephony codec/bandwidth and device mix expected in deployment. Do not train mostly on 16–48 kHz close-talk speech and assume downsampling reproduces telephone artifacts. Metadata should include codec, packet loss if applicable, mobile/landline/VoIP, and channel side.

7.5 Noisy industrial / in-car speech

Synthetic mixing is useful only if the nuisance library is itself diverse. Andrew Ng's car-noise example emphasizes a subtle failure mode: repeatedly mixing 1,000 h of speech with only one hour of car noise can make a huge synthetic corpus that overfits a narrow noise source [42]. SpecAugment provides a complementary feature-domain augmentation strategy and has proved effective from public benchmarks to large multidomain datasets [39, 44].

7.6 Code-switching

Collect *natural* switches rather than only concatenating monolingual utterances. Keep token/span language labels, switching context, speaker language proficiency where ethical and useful, and evaluate both overall and per-language errors. TALCS is a strong Mandarin-English example derived from real teaching interactions [18].

7.7 Low-resource languages

The lowest-cost path is usually: small gold corpus → strong multilingual/self-supervised model → pseudo-label/active selection → human correction → re-train. Recent systematic reviews identify recurrent obstacles in orthographic instability, speaker recruitment, infrastructure, annotation cost, licensing, and reproducibility [36, 37]. Community governance and orthography decisions must happen before massive recording, because post-hoc normalization can destroy linguistic information.

8 Transcription and annotation architecture

A robust corpus should not force every downstream system to use the same text representation. Store multiple synchronized layers.

Table 3: Recommended annotation layers.

Layer	Purpose	Typical fields
Raw/verbatim transcript	Auditability and spontaneous-speech fidelity	exact words, fillers, partials, repetitions, hesitations
Normalized orthography	Stable ASR target	punctuation/casing policy, numbers, abbreviations, spacing
Pronunciation/reading	Language-specific QA and alignment	phones, Pinyin+tone, Korean pronunciation form, Japanese reading
Time alignment	Training and retrieval	utterance, word, or phone start/end times; alignment confidence
Speaker/turn layer	Diarization and leakage control	pseudonymous speaker ID, turn ID, overlap flags
Acoustic metadata	Robustness analysis	sample rate, device, distance, SNR, reverberation/quality score
Linguistic metadata	Coverage and fairness	BCP-47 language, region, dialect/accent, code-switch spans
Governance/provenance	Legal reproducibility	source URL/ID, license, consent basis, collection date, model/version used for pseudo-labeling

Do not overwrite raw transcripts during normalization. Treat normalization as a deterministic, versioned transform:

$$y_{\text{norm}} = f_{\theta,v}(y_{\text{raw}}), \tag{2}$$

where v is the normalization-policy version. This makes WER/CER comparisons reproducible when policies change.

9 Segmentation, alignment, and automatic filtering

Long recordings should be converted into utterance-scale training examples without destroying context needed for future tasks. A typical automated pipeline is:

$$\text{source media} \rightarrow \text{VAD/diarization} \rightarrow \text{candidate transcript} \rightarrow \text{alignment} \rightarrow \text{quality filters} \rightarrow \text{human audit}. \quad (3)$$

GigaSpeech uses forced alignment and segmentation plus transcription-quality filtering [3]. WenetSpeech uses different candidate-generation mechanisms for video captions and podcasts, then applies label-error detection [13]. JTubeSpeech uses CTC-based methods to filter automatically collected Japanese YouTube data [20]. ReazonSpeech demonstrates bootstrapped TV alignment at very large scale [22]. WorldSpeech goes further by storing human transcript, ASR transcript, CER, SNR, quality scores, source, and time offsets, making quality and provenance queryable [12].

Recommended Store the score that caused a sample to pass a filter. A binary "accepted" flag alone destroys information needed to re-filter later.

10 Validation and quality assurance

Spoken-language-resource validation literature argues that quality is multi-dimensional: consistency, fitness for use, reusability, documentation, best-practice compliance, independent validation, and accessibility all matter [31]. A modern QA system should therefore track a vector rather than collapse everything into one score:

$$q_i = [\text{SNR},\ \text{MOS},\ \text{CER}_{align},\ \text{clip},\ \text{langConf},\ \text{speakerConf},\ \text{metadataCompleteness},\ \text{licenseState}]. \quad (4)$$

Recommended QA stages:

1. **Pre-validation:** validate prompt design, annotation schema, consent text, file naming, and a mini-database before large-scale collection.
2. **Online QA:** clipping, silence ratio, sample-rate checks, VAD sanity, duplicate audio, transcript character set, duration limits, language ID, and speaker/segment consistency.
3. **Alignment QA:** compare weak/ASR transcript to reference/caption and retain per-segment confidence/error.
4. **Human sampled QA:** listen to stratified samples from every source, language variety, device, and quality band; oversample borderline cases.
5. **Pre-release external validation:** verify documentation, speech/annotation links, corpus composition, transcript conventions, pronunciation resources, and split integrity [31].

The Slovenian ARTUR experience reinforces that legal management, precise word-by-word transcription/segmentation, and collaborator/workflow organization are major cost drivers, not peripheral administrative details [32].

11 Split construction: preventing silent evaluation leakage

The test set must represent future data, not merely unused files. Split at the highest dependency level available:

$$S_{train} \cap S_{test} = \varnothing, \qquad Session_{train} \cap Session_{test} = \varnothing, \quad (5)$$

with additional source-level isolation for found media. For example, two clips from the same YouTube video, audiobook chapter, meeting, broadcast episode, or repeated speaker can leak lexical, acoustic, and speaker information even when filenames differ.

Recommended split rules:

- **speaker-disjoint** for general speaker-independent ASR;
- **session-disjoint** to avoid repeated room/device/background;

- **source/show/channel-disjoint** for web/broadcast data;
- **time-disjoint** when robustness to future terminology/domain drift matters;
- **dialect-balanced challenge sets** for language-variety fairness;
- **domain-mismatched tests** in addition to a matched dev/test pair.

STT4SG-350's use of the same test sentences across Swiss German dialect regions is a particularly clean design for comparing regional performance [28]; WenetSpeech's matched Internet and mismatched meeting tests provide another useful pattern [13].

12 Data diversity and fairness are accuracy requirements

Speech datasets encode whose speech the system can recognize. Ethical dataset work argues for legal/privacy protection, targeted demographic sampling based on user needs, appropriate metadata, and situated design [34]. The augmented speech-datasheet study shows that dataset documentation should explicitly describe language/accent/dialect diversity, age/gender where collected, recording environment/device, license, consent, privacy, compensation, and whether the data are derivative [33].

A useful rule is: *if a subgroup is important enough to promise product performance, it is important enough to receive its own corpus coverage statistic and evaluation slice.* Do not infer fairness from global WER.

13 Representative corpus evidence

Table 4: Representative corpora and design lessons used in this synthesis. Hours are approximate values reported by the cited source/release.

Corpus	Lang.	Hours	Collection mode	Notable design choices	Lesson for corpus construction
LibriSpeech [1]	English	1,000	Audiobooks	16 kHz read speech, public-domain source, LM resources	Clean read/found data can create a reproducible benchmark, but domain is narrow.
GigaSpeech [3]	English	10,000 labeled / 40,000 total	Audiobooks, podcasts, YouTube	Forced alignment/segmentation, quality caps, professional dev/test transcription	Scale is viable when alignment filtering and protected evaluation are explicit.
People's Speech [4]	English	30,000	Licensed web audio	Searches for appropriately licensed audio with existing transcripts	Legal usability can be a corpus-design objective, not an afterthought.
SPGISpeech [5]	English	5,000	Earnings calls	Professional transcripts, fully formatted targets	Domain-specific high-quality data can be more valuable than generic scale.
SpeechStew [6]	English	multi-corpus	Mixture	Meetings, broadcast, CV, LibriSpeech, Switchboard/Fisher, TED-LIUM, WSJ	Heterogeneous domain mixture improves transfer and robustness.
AISHELL-1 [14]	Mandarin	∼170	Recorded read speech	Detailed capture devices/environments, lexicon, public recipes	Document acquisition and linguistic resources alongside audio.
AISHELL-2 [15]	Mandarin	1,000	Read speech	iOS train; Android/mic dev/test	Cross-channel testing exposes device mismatch.
KeSpeech [16]	Mandarin + subdialects	1,542	Large speaker recording	27,237 speakers, 34 cities, 8 subdialects + labels	Geographic/speaker diversity should be designed, measured, and labeled.
WenetSpeech [13]	Mandarin	22,400 total	YouTube/podcast	10k+ h labeled, 2.4k+ weak, 10k unlabeled; OCR/ASR candidates; manual tests	Tiered quality and matched/mismatched tests are strong modern patterns.
MagicData-RAMC [17]	Mandarin	180	Mobile conversations	15 domains, manual transcript, speaker VAD times, metadata	Rich annotations enable ASR + diarization + domain analysis.
TALCS [18]	Mandarin-English	587	Real online teaching	Natural code-switching, 16 kHz, documented devices/environments	Natural code-switch data is preferable to artificial concatenation for realistic switching.
KsponSpeech [19]	Korean	969	Two-person open-domain dialogue	2,000 speakers; orthography+pronunciation; disfluency tags	Language-specific transcript layers matter; audit demographic imbalance.
JNAS [23]	Japanese	60	Read speech	306 speakers balanced by sex, newspaper + phonetically balanced prompts	Prompt design can ensure phonetic/text coverage efficiently.

Corpus	Lang.	Hours	Collection mode	Notable design choices	Lesson for corpus construction
CSJ [24]	Japanese	650+	Spontaneous speech	Broad corpus plus richly annotated 44 h-ish Core	Concentrate expensive deep annotation on a representative core.
JTubeSpeech [20]	Japanese	1,300+	YouTube + subtitles	CTC-based automatic filtering	Language-light automatic filtering can expand non-English data.
Japanese TV corpus [21]	Japanese	2,000+	TV + subtitles	Iterative lightly supervised alignment	Iterative alignment can convert captions into useful ASR pairs.
ReazonSpeech [22]	Japanese	15,735 initial	TV + captions	Bootstrapped labeling; open creation toolkit; later releases much larger	Corpus creation can be a continuously evolving pipeline rather than a static release.
Tuda-De + Spoken Wikipedia [25]	German	412 combined	Read + crowd/wiki	Conservative vs minimal alignment pruning; multiple microphones in Tuda-De	Alignment filtering and source mixing improve open German ASR.
ASR Bundestag [26]	German	610 labeled + 1,038 unlabeled	Parliament	Automated alignment, supervised + SSL tiers	Public institutional speech is a strong domain source with natural vocabulary.
SDS-200 [27]	Swiss German	200	Web crowdsourcing	∼4,000 speakers; dialect/age/gender; peer validation	Crowd validation plus dialect metadata can scale a non-standardized variety.
STT4SG-350 [28]	Swiss German	343	Web recording	balanced dialect-region amounts; gender balance; all ages; same test sentences per region	Balanced challenge design enables interpretable fairness comparisons.
WorldSpeech [12]	76+ langs.	65,000	Public institutional / broadcast / audiobook	Human transcript + ASR transcript + CER + SNR/quality + provenance; iterative realignment	Keep quality/provenance features as first-class data and iterate alignment.
VoxPopuli [9]	23 langs.	100,000 unlabeled	European Parliament	1.8k h transcribed subset plus interpretation data	Large unlabeled pools and smaller labeled layers support SSL/semi-supervised learning.
FLEURS [10]	102 langs.	∼12/lang	Parallel prompted speech	n-way parallel across languages	Parallel benchmark content enables cross-language comparability.
MMS [11]	1,000+ langs.	massive multilingual	Public readings + SSL	1,406-language pretraining, 1,107-language ASR	For long-tail languages, breadth plus SSL can outweigh conventional per-language scale.

14 Recommended corpus schema

A practical JSONL manifest should make provenance and multiple transcript layers explicit. One example is:

```
{
  "utterance_id": "ko_000001",
  "audio": "audio/spk042/ko_000001.flac",
  "duration_s": 4.62,
  "sample_rate": 16000,
  "speaker_id": "spk042",
  "session_id": "sess_2026_09_001",
  "language": "ko-KR",
  "region": "KR-11",
  "dialect": "standard",
  "text_raw": "...",
  "text_normalized": "...",
  "pronunciation": "...",
  "code_switch_spans": [],
  "start_s": 102.31,
  "end_s": 106.93,
  "snr_db": 21.4,
  "quality_score": 3.7,
  "transcript_source": "human_verified",
  "alignment_cer": 0.01,
  "source_id": "fieldrec_001",
  "license": "CC-BY-4.0",
  "consent_version": "v2.1",
  "split": "train"
}
```

For found media, add `source_url`, `source_start_s`, `source_end_s`, caption/ASR model version, and an immutable provenance hash. For crowdsourced data, retain collection-app version and validation-vote metadata. For pseudo-labels, store model/checkpoint, decoding parameters, confidence, and the date generated.

15 Text normalization by language

Table 5: Recommended text layers and normalization focus.

Language	Preserve	Normalize/audit
English	raw casing/punctuation if available; contractions/slang	numbers, currency, units, acronyms, URLs, filled pauses, punctuation policy, named entities
German	canonical orthography incl. umlauts/ß; standard case	compounds, abbreviations, numbers, de-DE/de-AT/de-CH variants, optional lowercase training view
Korean	raw Hangul, normalized orthography, pronunciation layer	spacing, counters/numbers, sound-change representation, disfluency tags, foreign terms
Mandarin	Hanzi, normalized text, pronunciation metadata	Pinyin+tone QA, numbers/classifiers, regional pronunciations, English spans, punctuation
Japanese	surface text, normalized text, reading/morphology layer	kanji readings, numerals/counters, katakana/loanwords, long vowels/geminates, tokenization version

16 Data selection: more hours versus more information

Large pools contain redundancy. Once a basic corpus exists, new samples should be selected for expected information gain, target-distribution coverage, or model uncertainty rather than only randomly. A simple selection score can combine:

$$s(x) = \alpha\, U(x) + \beta\, C_{target}(x) + \gamma\, R_{rare}(x) - \delta\, Redundancy(x) - \eta\, Risk(x), \tag{6}$$

where U is model uncertainty, C_{target} is target-domain/embedding coverage, R_{rare} rewards underrepresented phones/words/accents/conditions, and $Risk$ includes poor alignment or licensing uncertainty. The principle is consistent with classical resource-scarce corpus selection and modern large-pool filtering [3, 13, 30].

17 Augmentation and synthetic data

Augmentation should model nuisance factors that are expected at deployment while preserving label semantics. Recommended augmentation categories include:

- speed perturbation and mild tempo variation;
- reverberation using diverse room impulse responses;
- additive noise using a *large and diverse* noise library;
- codec/telephone/channel simulation;
- SpecAugment in the spectrogram domain [39];
- TTS/self-training only when synthetic artifacts are measured and the evaluation set remains real.

Synthetic data should not replace rare real conditions entirely. If a system must recognize elderly, child, atypical, dialectal, or highly accented speech, generate augmentation only as a supplement to consented real speech from those populations. Otherwise, the model may learn a simulator rather than the target population.

18 Legal, ethical, and documentation requirements

Speech can be personally identifying, and public availability does not automatically imply ethically unrestricted reuse. Dataset design should specify:

- legal basis/license for recording, transcription, redistribution, derivative features, and model training;
- informed consent and withdrawal/deletion process for newly recorded participants;
- treatment of personally identifiable information in both audio and transcript;
- whether demographic/dialect metadata are self-reported, inferred, or externally assigned;
- compensation and crowdworker conditions where relevant;
- intended use, prohibited use, and known population/domain limitations;
- maintenance/version policy and procedures for correcting labels after release.

Papakyriakopoulos et al. emphasize that ethical speech data requires diversity, inclusion, privacy, data-subject protection, and documentation rather than a single one-size-fits-all template [33, 34]. ARTUR demonstrates that copyrights, personality rights, and personal-data protection are among the main operational constraints in building open corpora [32].

19 A practical “best corpus” blueprint

For a new general-purpose ASR effort, the following staged process is recommended.

Phase 0 - Product and evaluation specification

Define languages/varieties, users, acoustic environments, speech styles, domains, latency mode, output orthography, and failure costs. Freeze an evaluation datasheet before large-scale collection.

Phase 1 - 20–50 h pilot gold set

Record/collect diverse speakers and environments; manually transcribe; establish normalization and pronunciation rules; validate data formats and metadata; build an initial model and error taxonomy. For highly constrained tasks, this may already produce a useful seed system [30].

Phase 2 - 50–200 h gold core

Fill coverage gaps discovered in Phase 1. Make speakers/sessions disjoint. Add dialect/accent challenge sets. Deeply annotate a subset for alignment, pronunciation, speaker turns, overlap, and error analysis. This is the layer that should survive model generations.

Phase 3 - 200–1,000+ h validated supervised expansion

Use ASR-assisted transcription, forced/CTC alignment, and aggressive filtering. Human-audit every source/stratum and the confidence tail. WorldSpeech suggests that, with a strong pretrained model, the first 200 h can provide especially large adaptation gains in many languages, with smaller marginal gains after roughly 500 h in its tested setting [12].

Phase 4 - 1,000–100,000+ h weak/unlabeled reservoir

Harvest legally usable web/broadcast/institutional speech. Store provenance, captions, pseudo-label model version, alignment error, SNR, and quality scores. Use self-supervised pretraining, semi-supervised training, or iterative re-alignment. GigaSpeech, WenetSpeech, VoxPopuli, Whisper-style weak supervision, MMS, and ReazonSpeech show multiple successful variants of this pattern [3, 7, 9, 11, 13, 22].

Phase 5 - Continuous error-driven collection

Production errors become a data acquisition queue. Every release should report WER/CER by language variety, environment, device, speaking style, SNR, and other relevant slices; the next data round targets the worst high-priority slices rather than blindly adding hours.

20 Common corpus-construction failures

1. **Hours as the only KPI.** Ten thousand redundant clean read hours may underperform a smaller deployment-matched mixture.
2. **Speaker leakage.** Random utterance splitting can put the same speaker/session in train and test.
3. **Found-media leakage.** Multiple clips from the same episode/channel/book can contaminate evaluation.
4. **Silent normalization.** Overwriting raw transcript text makes annotation decisions irreversible.
5. **Treating language names as homogeneous.** “English,” “German,” “Korean,” “Mandarin,” and “Japanese” all hide regional, social, age, domain, and style variation.
6. **Weak labels without provenance.** A pseudo-label is not a gold label; store model/version/confidence and keep it separable.
7. **Synthetic diversity from narrow sources.** Repeating the same noise/room/voice transformations can create the illusion of scale while reducing real diversity [42].
8. **One aggregate WER.** Global averages can hide serious subgroup failures [38].
9. **Licensing late in the project.** A technically excellent corpus that cannot legally be redistributed or used in the intended product may have little practical value [4, 32].
10. **No dataset maintenance.** Large evolving corpora require versioning, correction logs, de-duplication, and stable evaluation releases.

21 Recommended metrics for the corpus itself

Model metrics alone are insufficient. Report corpus metrics before training:

- total hours and utterances by split;
- unique speakers and median/95th percentile hours per speaker;
- hours by language, region/dialect/accent, age band, gender if collected, device, environment, and domain;
- duration distribution, silence ratio, clipping rate, SNR/quality distributions;
- transcript vocabulary, character/subword coverage, OOV/rare-token coverage;
- proportion human-transcribed, human-verified, weakly labeled, pseudo-labeled, unlabeled;
- alignment error/confidence distribution;
- duplicate/near-duplicate rate;
- license/consent coverage and metadata completeness;
- leakage audit results across speaker, session, source, and text prompt.

For multilingual corpora, report these per language rather than only totals. The low-resource literature repeatedly shows that a corpus can look large globally while many individual languages remain severely underrepresented [10, 11, 36].

22 Conclusions

The evidence supports a shift from "collect as many hours as possible" to **data-centric corpus engineering**. The best ASR corpus is an evolving system with explicit target coverage, tiered label trust, reproducible normalization, protected evaluation, measurable quality, and documented governance.

For the five languages considered here:

- **English:** prioritize accent/sociolinguistic and domain diversity; use subgroup challenge sets rather than assuming abundant data implies fairness.
- **German:** explicitly model de-DE/de-AT/de-CH and regional varieties where relevant; preserve standard orthography and design for compounds; use dialect-balanced tests.
- **Korean:** preserve orthographic and pronunciation-oriented layers, enforce a consistent spacing policy, and annotate spontaneous disfluencies; balance age/region beyond capital-area young speakers.
- **Mandarin:** combine read, conversational, found-media, regional/subdialect, and code-switch data; retain tone-aware pronunciation metadata for audit even with character/subword models.
- **Japanese:** combine controlled read speech with spontaneous/broadcast data; preserve surface form plus reading/morphology; use a broad corpus plus richly annotated core.

Across all languages, a small trusted core should be treated as infrastructure, not merely training data. It defines what "correct" means, anchors evaluation, and enables new models to be compared fairly. Scale should then be added through validated and weakly labeled tiers with provenance and confidence retained. This architecture is more robust, auditable, and economically sustainable than either a tiny pristine corpus or a massive opaque scrape.

Limitations

This synthesis does not claim a new manual full-text review of 500 individual papers. It satisfies the requested large evidence scope by incorporating systematic evidence maps whose search frames collectively cover many thousands of publications, then grounding recommendations in directly inspected representative corpus and methodology papers. Search engines and mirror sites can omit inaccessible papers, and some very recent 2026 sources are preprints. Corpus

sizes can also change after publication (e.g., continuously evolving datasets); values in the representative table refer to the cited paper/release unless otherwise stated.

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